

Corporate Finance

Lecture 2 & 3: Optimal Portfolio Choice

Camilo Saldarriaga

Finance Department, ESCP Business School, 2025/2026

Introduction

- ▶ Optimal portfolio choice:
 - ▶ How can we optimise risk and return by combining multiple assets?
 - ▶ Pioneered by Markowitz in his article “Portfolio Selection” published in the “Journal of Finance” in 1952.



Outline

- A. Risk and return of a two-stock portfolio: the effect of diversification
- B. Efficient portfolios with two stocks
- C. N risky stocks: the efficient frontier
- D. The “super-efficient” portfolio
- ▶ The content of this lecture mostly draws on the BDM textbook Chap. 10.5-6 and 11.1-5.

BEFORE WE START

Which framework for choice under risk?

- ▶ Let's play a game
- ▶ Do you prefer:
 - A. Win EUR 100 000 for sure.
 - B. Participate in a lottery which yields EUR 0 with probability 0.5, and EUR 200 000 with probability 0.5?

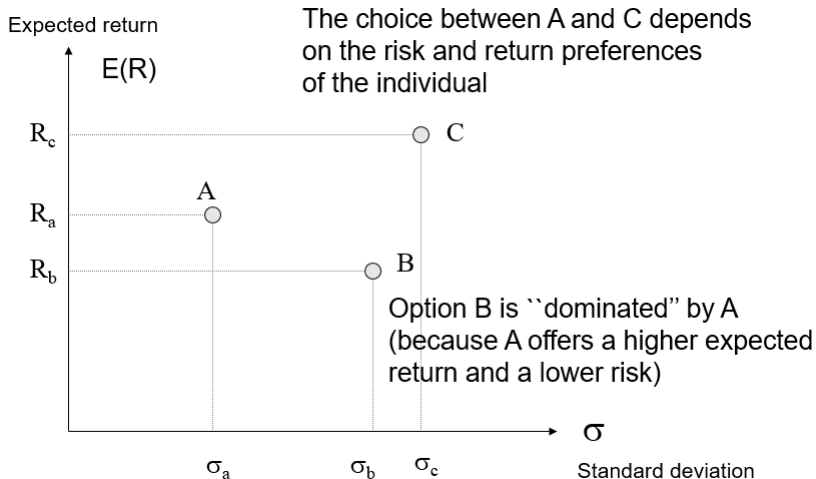
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 - ▶ $E(W_A) = 100000$
 - ▶ $E(W_B) = 0.5 \times 0 + 0.5 \times 200000 = 100000$

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- ▶ According to the expected value criterion: indifference.
- ▶ And still: (in general) preference for the sure outcome! (i.e. aversion to risk)

The mean-variance framework



Mean-variance utility

- ▶ Mean-variance analysis can be motivated with utility derived from the first two moments of wealth: $U(E(W), \text{Var}(W))$, where U is increasing in the expected wealth, $E(W)$, and decreasing in the variance of wealth, $\text{Var}(W)$.
- ▶ A common specification is

$$U(E(W), \text{Var}(W)) = E(W) - \frac{\gamma}{2} \text{Var}(W),$$

where $\gamma > 0$ measures the aversion to variance relative to the preference for the mean. γ is related to the coefficient of absolute risk aversion discussed in the previous lecture.

- ▶ In our simplified framework more risk aversion or lower risk tolerance refer to higher values of γ .

A. Risk and return of a two-stock portfolio: the effect of diversification

- ▶ You can invest in two stocks:
- ▶ EUR 50 invested in Umbrella Inc and EUR 50 in Sun Umbrella Inc:
- ▶ The appropriate combination of stocks can produce an investment with lower risk.

Two risky stocks: a more realistic example of diversification (with fictitious stocks)

Year	Stock Returns			Portfolio Returns	
	North Air	West Air	Tex Oil	$1/2R_N + 1/2R_W$	$1/2R_W + 1/2R_T$
2013	21%	9%	-2%	15.0%	3.5%
2014	30%	21%	-5%	25.5%	8.0%
2015	7%	7%	9%	7.0%	8.0%
2016	-5%	-2%	21%	-3.5%	9.5%
2017	-2%	-5%	30%	-3.5%	12.5%
2018	9%	30%	7%	19.5%	18.5%
Average Return	10.0%	10.0%	10.0%	10.0%	10.0%
Volatility	13.4%	13.4%	13.4%	12.1%	5.1%

Probability reminder

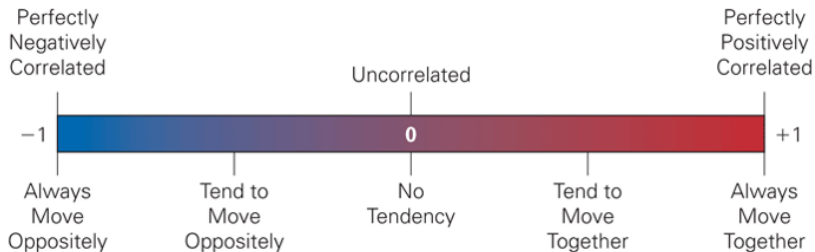
- Computation of the covariance and correlation: Consider the random variables X, Y with payments $x_i, y_i, i = 1, \dots, N$:

$$\begin{aligned}\text{Covariance}(X, Y) &= \sigma_{X,Y} = E[(X - E(X))(Y - E(Y))], \\ &= \sum_i p_i (x_i - E(X))(y_i - E(Y)).\end{aligned}$$

$$\text{Correlation}(X, Y) = \rho_{X,Y} = \frac{\text{Covariance}(X, Y)}{\sqrt{\text{Variance}(X) \text{Variance}(Y)}} = \frac{\sigma_{X,Y}}{\sigma_X \sigma_Y}.$$

$$\text{Beta}_X^Y = \beta_{X,Y} = \frac{\text{Covariance}(X, Y)}{\text{Variance}(Y)} = \frac{\sigma_{X,Y}}{\sigma_Y^2}.$$

Interpretation of the correlation



The correlation measures how returns move in relation to each other.

Computing the covariance and correlation

- Computation of the covariance with historical returns

$$\text{Cov}(R_i, R_j) = \frac{1}{T-1} \sum_{t=1}^T (R_{it} - \bar{R}_i) (R_{jt} - \bar{R}_j).$$

- The correlation is

$$\text{Corr}(R_i, R_j) = \frac{\text{Cov}(R_i, R_j)}{\sigma_{R_i} \sigma_{R_j}}.$$

- **Reminder:** computation of expected returns and variance from historical data

$$\bar{R} = \frac{1}{T} \sum_{t=1}^T R_t = \frac{1}{T} (R_1 + \dots + R_T),$$

$$\text{Var}[R] = \frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})^2.$$

Computing the covariance and correlation between two stocks

Year	Deviation from Mean			North Air and West Air	West Air and Tex Oil
	$(R_N - \bar{R}_N)$	$(R_W - \bar{R}_W)$	$(R_T - \bar{R}_T)$	$(R_N - \bar{R}_N)(R_W - \bar{R}_W)$	$(R_W - \bar{R}_W)(R_T - \bar{R}_T)$
2013	11%	-1%	-12%	-0.0011	0.0012
2014	20%	11%	-15%	0.0220	-0.0165
2015	-3%	-3%	-1%	0.0009	0.0003
2016	-15%	-12%	11%	0.0180	-0.0132
2017	-12%	-15%	20%	0.0180	-0.0300
2018	-1%	20%	-3%	-0.0020	-0.0060
Sum = $\sum_i (R_{i,t} - \bar{R}_i)(R_{j,t} - \bar{R}_j) =$				0.0558	-0.0642
Covariance:	$Cov(R_i, R_j) = \frac{1}{T-1} \text{Sum} =$			0.0112	-0.0128
Correlation:	$Corr(R_i, R_j) = \frac{Cov(R_i, R_j)}{SD(R_i)SD(R_j)} =$			0.624	-0.713

Historical annual volatilities and correlations for selected stocks

	Microsoft	HP	Alaska Air	Southwest Airlines	Ford Motor	Kellogg	General Mills
Volatility (Standard Deviation)	32%	36%	36%	31%	47%	19%	17%
Correlation with Microsoft	1.00	0.40	0.18	0.22	0.27	0.04	0.10
HP	0.40	1.00	0.27	0.34	0.27	0.10	0.06
Alaska Air	0.18	0.27	1.00	0.40	0.15	0.15	0.20
Southwest Airlines	0.22	0.34	0.40	1.00	0.30	0.15	0.21
Ford Motor	0.27	0.27	0.15	0.30	1.00	0.17	0.08
Kellogg	0.04	0.10	0.15	0.15	0.17	1.00	0.55
General Mills	0.10	0.06	0.20	0.21	0.08	0.55	1.00

Properties of the covariance and variance (reminder)

- The covariance is bilinear:

$$\text{Cov}[R_1 + R_2, R_3] = \text{Cov}[R_1, R_3] + \text{Cov}[R_2, R_3]$$

$$\text{Cov}[mR_2, R_3] = m\text{Cov}[R_2, R_3]$$

- The variance is a special case of the covariance

$$\text{Var}[R_1] = \text{Cov}[R_1, R_1]$$

$$\text{Var}[mR_1] = m\text{Cov}[R_1, mR_1] = m^2\text{Cov}[R_1, R_1] = m^2\text{Var}[R_1]$$

Computing the variance and standard deviation of a portfolio: the case with two stocks

► The return of the portfolio is $R_p = x_1 R_1 + x_2 R_2$.

► We get

$$\begin{aligned}\text{Var}(R_p) &= \text{Cov}(R_p, R_p) = \text{Cov}(x_1 R_1 + x_2 R_2, x_1 R_1 + x_2 R_2) \\ &= x_1^2 \text{Var}(R_1) + x_2^2 \text{Var}(R_2) + 2x_1 x_2 \text{Cov}(R_1, R_2) \\ &= x_1^2 \text{Var}(R_1) + x_2^2 \text{Var}(R_2) + 2x_1 x_2 \sigma_{R_1} \sigma_{R_2} \text{Corr}(R_1, R_2)\end{aligned}$$

► The volatility of the portfolio is

$$\sigma_{R_p} = \sqrt{\text{Var}(R_p)}.$$

The risk of a portfolio with two stocks

- ▶ The risk of a portfolio with two stocks is **not** the sum of the individual stock risks.
- ▶ It is the sum of the individual stock risks, plus the comovement of the stocks:

$$\text{Var}(R_p) = \underbrace{x_1^2 \text{Var}(R_1) + x_2^2 \text{Var}(R_2)}_{\text{individual risks}} + \underbrace{2x_1 x_2 \text{Cov}(R_1, R_2)}_{\text{comovement}}$$

- ▶ If they co-move in different directions, the aggregated risk is smaller.
- ▶ If they co-move in the same direction, the aggregated risk is greater.

Diversification and correlation

$\rho_{12} = 1 \Rightarrow \sigma_p = x_1\sigma_1 + x_2\sigma_2$ The risk of the portfolio is the average risk.

$\rho_{12} < 1 \Rightarrow \sigma_p < x_1\sigma_1 + x_2\sigma_2$ The risk of the portfolio is smaller than the average risk due to the effect of diversification.

$\rho_{12} = -1 \Rightarrow \sigma_p = |x_1\sigma_1 - x_2\sigma_2|$ Portfolio weights x_1 and x_2 exist which eliminate the risk.

The volatility of a portfolio of N stocks

- The portfolio return is

$$R_p = x_1 R_1 + \cdots + x_N R_N = \sum_{i=1}^N x_i R_i.$$

- The variance is

$$\begin{aligned}\text{Var} [R_p] &= \text{Cov} [R_p, R_p] = \sum_{i=1}^N x_i \text{Cov} [R_i, R_p], \\ &= \sum_{i=1}^N x_i \text{Cov} \left[R_i, \sum_{j=1}^N x_j R_j \right] = \sum_{i=1}^N x_i \sum_{j=1}^N x_j \text{Cov} [R_i, R_j] \\ &= \sum_{i=1}^N \sum_{j=1}^N x_i x_j \text{Cov} [R_i, R_j].\end{aligned}$$

- **Don't panic:** We will not use this formula for more than two stocks.

B. Efficient portfolios with two stocks

Stock	Expected return	Volatility
Intel	26%	50%
Coca-Cola	6%	25%

- ▶ We assume that the correlation between the two stocks is zero.
 - ▶ Which portfolio should we choose?

Efficient portfolios with two stocks (con't 2)

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- ▶ Example: portfolio invested to 40% in Intel and 60% in Coca-Cola.

Efficient portfolios with two stocks (con't 2)

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$$E[R_{40-60}] = x_I E[R_I] + x_C E[R_C] = 0.4 \times 0.26 + 0.6 \times 0.06 = 0.14$$

$$\begin{aligned}\text{Var}[R_{40-60}] &= x_I^2 \text{Var}[R_I] + x_C^2 \text{Var}[R_C] + 2x_I x_C \text{Corr}(R_I, R_C) \sigma_I \sigma_C \\ &= 0.4^2 \times 0.5^2 + 0.6^2 \times 0.25^2 + 0 \\ &= 0.0625\end{aligned}$$

$$\sigma_{40-60} = \sqrt{0.0625} = 25\%$$

Efficient portfolios with two stocks (con't 3)

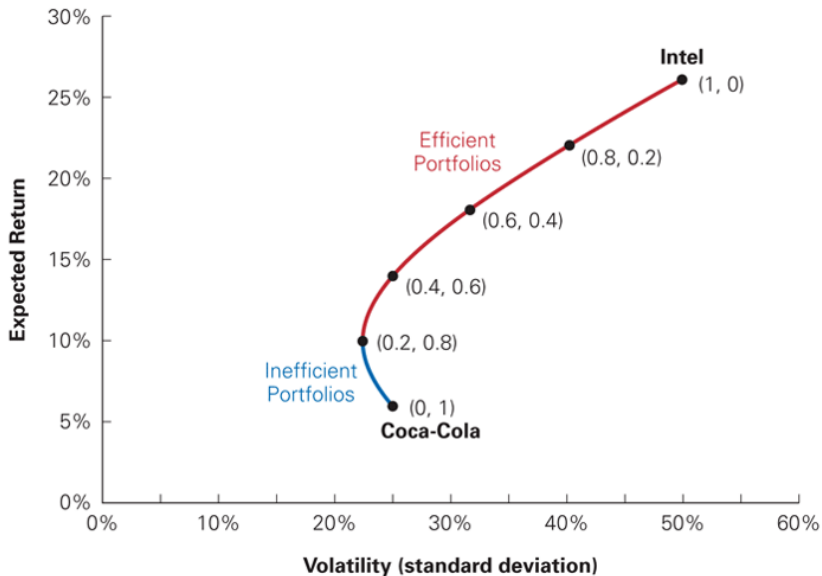
Portfolio Weights		Expected Return (%)	Volatility (%)
x_I	x_C	$E[R_p]$	$SD[R_p]$
1.00	0.00	26.0	50.0
0.80	0.20	22.0	40.3
0.60	0.40	18.0	31.6
0.40	0.60	14.0	25.0
0.20	0.80	10.0	22.4
0.00	1.00	6.0	25.0

Expected returns and volatility for different portfolios of two stocks.

But wait... “an efficient portfolio” ’ ?

- ▶ We define as efficient, all portfolios with:
 - ▶ Minimum variance for a given expected return;
 - ▶ Maximum expected return for a given variance.
- ▶ A portfolio is inefficient when it is possible to find another portfolio which has a larger expected return and a lower or equal volatility.
- ▶ Hence: investors who simultaneously maximise the expected return on their portfolio and minimise risk, will choose an efficient portfolio.

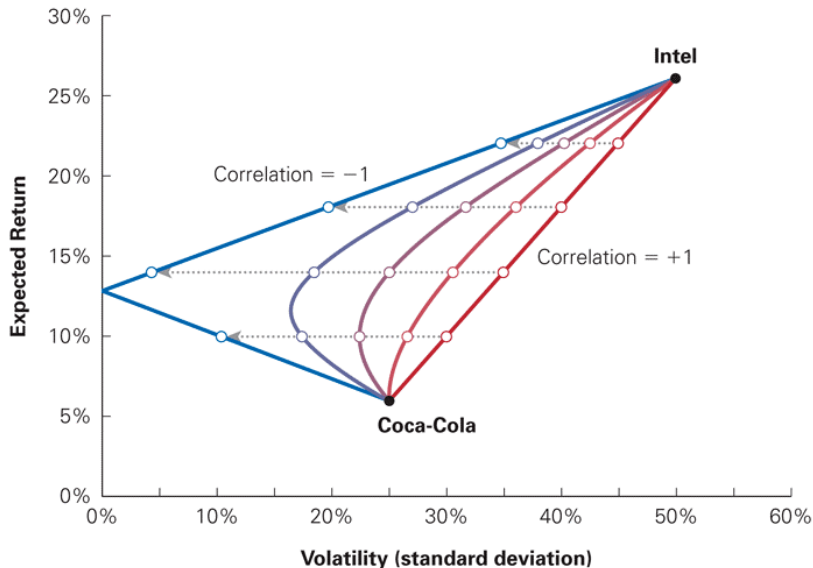
Back to our example



Improving returns with an efficient portfolio: the effect of correlation

- ▶ Consider a portfolio which consists to 100% of Coca-Cola stocks
- ▶ How can you maximize the expected return of this portfolio without increasing its volatility?

Improving returns with an efficient portfolio: the effect of correlation (con't)



The minimum variance portfolio

- It is easy to find the minimum variance portfolio in the case of two stocks:

$$\min_x \sigma_p^2 = x^2 \sigma_1^2 + (1-x)^2 \sigma_2^2 + 2(1-x)x \sigma_1 \sigma_2 \rho_{1,2}$$

The first-order condition is

$$2x\sigma_1^2 - 2(1-x)\sigma_2^2 + 2\sigma_1\sigma_2\rho_{1,2} - 4x\sigma_1\sigma_2\rho_{1,2} = 0$$

$$\Leftrightarrow x = \frac{\sigma_2^2 - \sigma_1\sigma_2\rho_{1,2}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_1\sigma_2\rho_{1,2}}.$$

Short sales

- ▶ Short sales make it possible to benefit from falling stock prices.
→ Strategy:
 - ▶ Take a short position: borrow and sell a stock.
 - ▶ Close a short position: buy the stock back and return it.
- ▶ Usually done when you expect the price to go down.

Short sales



Space Exploration Technologies Corp

NASDAQ: SPCX

Short sales

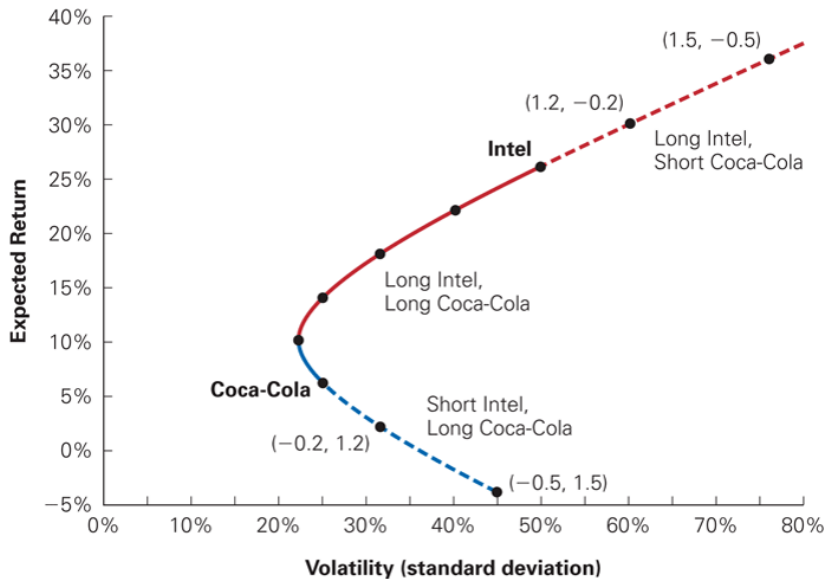
- ▶ Example: short sale of Stock A for a value of 100 and investment of the proceeds at the risk-free rate of 5% for one year. Consider the following scenarios at the end of the year:

Return of Stock A	10%	0%	-10%	-20%
Position in risk-free asset	105	105	105	105
Short position	-110	-100	-90	-80
Gain/loss	-5	5	15	25

- ▶ Mathematically, a short position corresponds to a negative portfolio weight in the stock.

Return of a portfolio with short sales

- ▶ Jérôme has 20 000 EUR available to invest for one year.
- ▶ He decides to sell short for a total of 10 000 EUR Coca-Cola stocks to buy Intel stocks for a total of 30 000 EUR
 - ▶ Remark: we still assume that the correlation between the two stocks is zero.
- ▶ What is the expected return and the volatility of his portfolio?

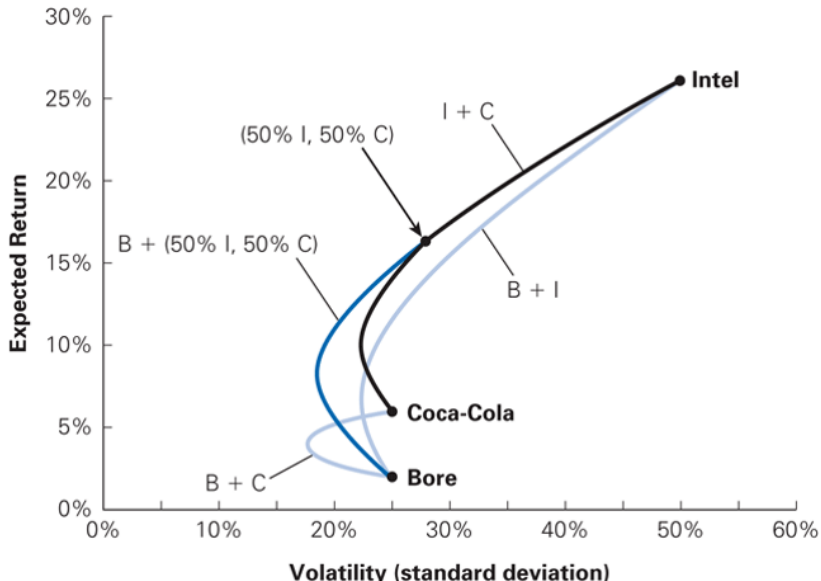


C. N risky stocks: the efficient frontier

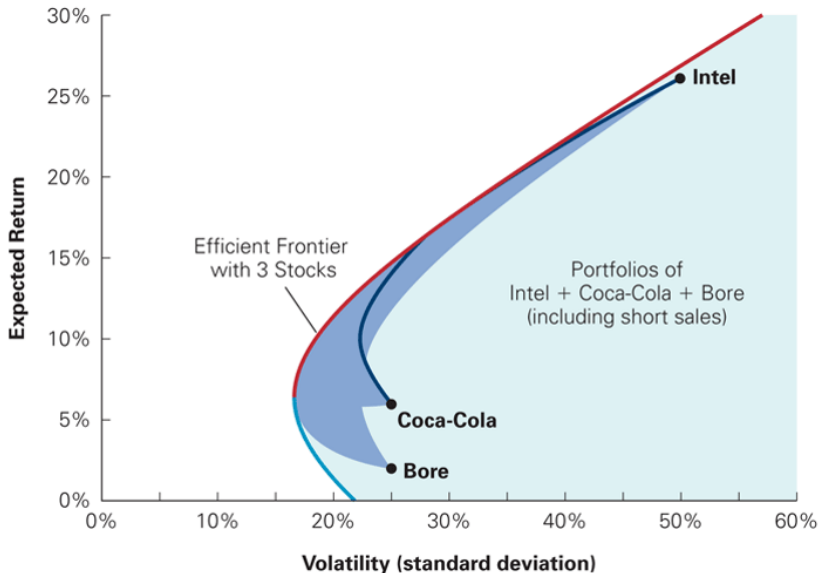
- ▶ What is the effect of adding a third stock to a portfolio of Coca-Cola and Intel?
- ▶ We assume that the stock of Bore Industries is uncorrelated with the other two stocks and that its expected return is only 2% and its volatility is the same as the one of the Coca-Cola stock (25%).
- ▶ Reminder: You would have to use this formula:

$$\begin{aligned}\text{Var} [R_p] &= \text{Cov} [R_p, R_p] = \sum_{i=1}^N x_i \text{Cov} [R_i, R_p], \\ &= \sum_{i=1}^N x_i \text{Cov} \left[R_i, \sum_{j=1}^N x_j R_j \right] = \sum_{i=1}^N x_i \sum_{j=1}^N x_j \text{Cov} [R_i, R_j] \\ &= \sum_{i=1}^N \sum_{j=1}^N x_i x_j \text{Cov} [R_i, R_j].\end{aligned}$$

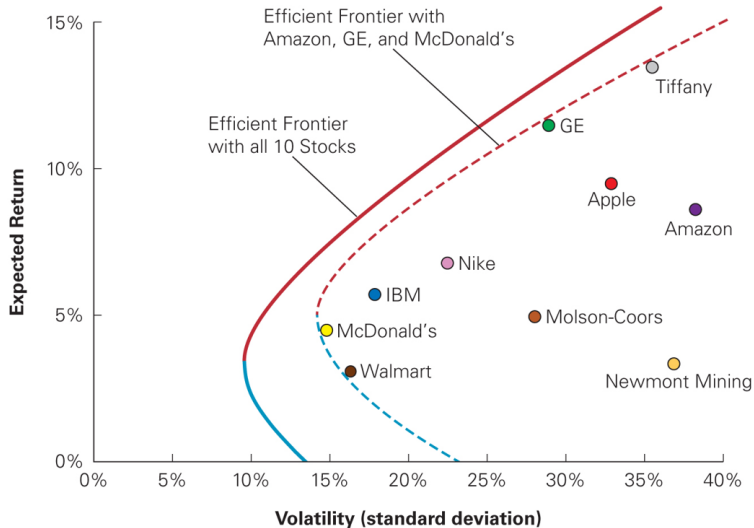
Expected return and volatility for selected portfolios of Intel, Coca-Cola, and Bore Industries stocks



Volatility and expected return for all portfolios of Intel, Coca-Cola, and Bore stock



Efficient frontier with three versus ten stocks



Summary on efficient portfolios

- ▶ We define as efficient, any portfolio with minimum variance for a given expected return.
 - ▶ Or: any portfolio with maximum return for a given level of risk.
- ▶ The efficient frontier is then the set of efficient portfolios.
 - ▶ It can be shown that it is the upper part of a hyperbola in the $(E(R), \sigma)$ plane.
- ▶ Augmenting the number of securities included in a portfolio allows to obtain an efficient portfolio with a lower risk, for a given level of expected return.
- ▶ Remark: an efficient portfolio may contain some short positions.

D. The “super-efficient” portfolio

- ▶ Up to now, the efficient portfolio only contained risky securities.
 - ▶ The investment decision depends on the preferences (“risk vs return”) of each investor
- ▶ Let us allow investors to invest in a risk-free asset
 - ▶ Its return is denoted by r_f (risk-free rate)
 - ▶ Zero risk

Risky asset and risk-free investment: the effect of leverage (James Tobin, 1958)

Two investments are available:

► Stocks of Umbrella Inc:

Return		Expected return	St.Dev.
Sun 1-2 -10%	Rain 30%	10%	20%

► A risk-free bank deposit

Return		Expected return	St.Dev.
Sun 1-2 3%	Rain 3%	3%	0%

Risky asset and risk-free investment: the effect of leverage (con't)

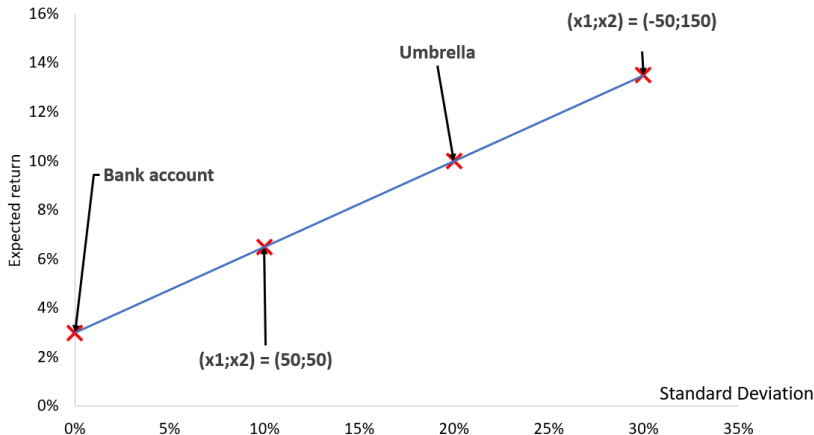
Invest 50 in Umbrella Inc and 50 in the bank account:

Risky asset and risk-free investment: the effect of leverage (con't 2)

Borrow 50 and invest 150 in Umbrella Inc:

Risky asset and risk-free investment: the effect of leverage (con't 3)

Possible combinations in the risk-return diagram



Leverage effect: general formula for the expected portfolio return and risk

- Expected return: weighted average (by the fraction invested) of each constituent's return:

$$E(R_p) = \sum_{i=1}^N x_i \times E(R_i) + x_f r_f, \text{ where assets } i = 1, \dots, N \text{ are risky and } x_f + \sum_{i=1}^N x_i = 1.$$

- Risk:

If $N = 1$, then $\sigma_p = x\sigma$, where x is the fraction invested in the risky asset and σ is its volatility.

If $N > 1$, then it suffices to calculate portfolio variance using only the risky assets with

$$\sigma_p^2 = \sum_{i=1}^N x_i^2 \sigma_i^2 + \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N x_i x_j \text{Cov}(R_i, R_j),$$

and take $\sigma_p = \sqrt{\sigma_p^2}$.

Leverage effect: consequences

The possibility to change the risk and return of a portfolio with a risk-free security has two fundamental consequences.

1. It is easy to generate high returns ...if a high level of risk is taken. $\Rightarrow$ Never analyse a return without also analysing its risk.
2. It is now possible to select objectively between two risky investments: $\Rightarrow$ The portfolio which generates the highest slope in the $(E(R), \sigma)$ plane is preferred, **regardless of the risk aversion**.

Sharpe ratio: identification of the best investment

- ▶ Sharpe ratio:

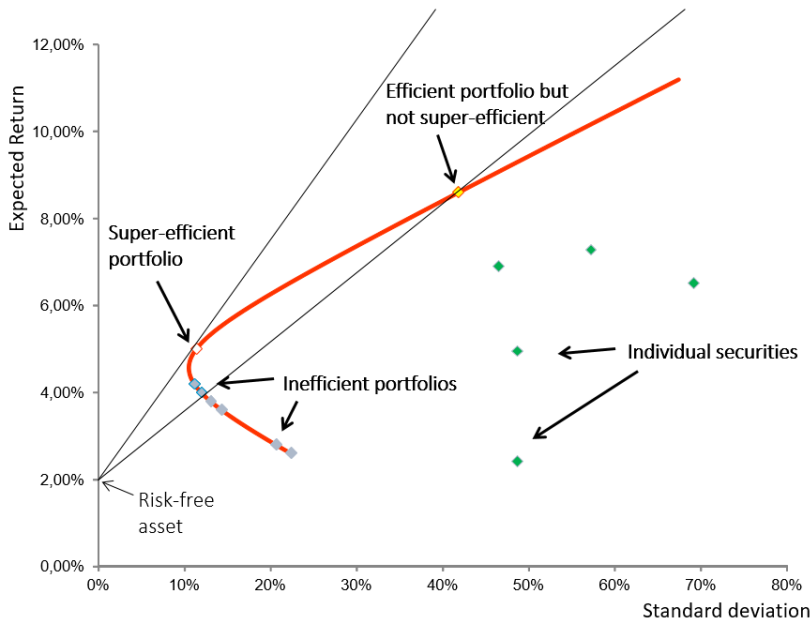
$$\frac{\text{Excess return of the portfolio}}{\text{Portfolio volatility}} = \frac{E[R_p] - r_f}{\sigma_p}.$$

- ▶ **Remark:** Here, the best investment does not depend on the aversion to risk. $\Rightarrow$ Regardless of the risk aversion, all investors should hold the portfolio of risky assets

What is the link between leverage, diversification, and efficient portfolios?

- ▶ If investors can invest and borrow at the risk-free rate, only one risky portfolio is efficient: the “tangency portfolio” or “super-efficient portfolio.”
- ▶ Conclusion:
 - ▶ From a given set of risky securities, investors should construct the same portfolio.
 - ▶ They will then adjust for their risk-aversion by investing/borrowing at the risk-free rate. $\Rightarrow$ Two fund separation theorem (James Tobin)
- ▶ It can then be shown that the efficient frontier is a half straight line, tangent to the Markowitz's hyperbola. The tangency point is the ...tangency portfolio.

The efficient frontier with a risk-free asset



Summary on efficient frontier with a risk-free asset

- It can be shown that the equation of the efficient frontier WITH risk-free asset (e.g., with borrowing/leverage) is given by:

$$E(R_p) = r_f + \underbrace{\frac{E(R_T) - r_f}{\sigma_T}}_{\text{Risk premium}} \sigma_p,$$

where the subscript p corresponds to the efficient portfolio and the subscript T to the tangency (or “super-efficient”) portfolio

- **Reminder:** all efficient portfolios contain the tangency portfolio and the risk-free asset (long or short)

Two types of risk

1. **Systematic risk**: common to all securities.



2. **Idiosyncratic risk**: specific to one security.

Diversification and risk

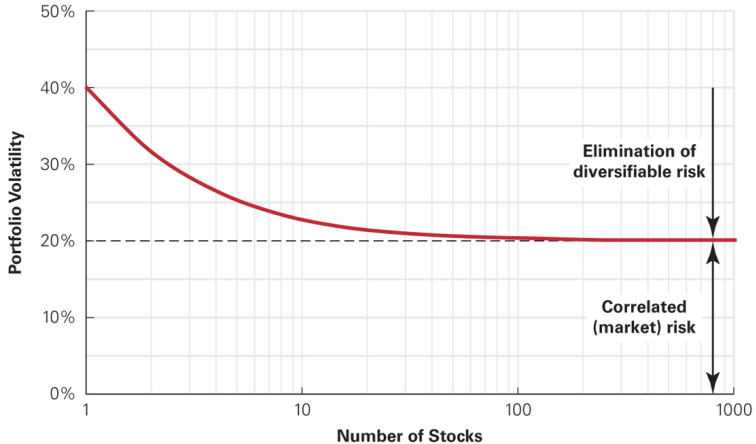
- ▶ Which risk do investors reduce by diversifying their portfolios?
 - ▶ The **systematic** risk? [risk common to all securities]
 - ▶ The **idiosyncratic** risk? [risk specific to a particular security]
- ▶ Remark: the variance of any portfolio can be written as

$$\sigma_p^2 = \sum_{i=1}^N x_i^2 \sigma_i^2 + \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N x_i x_j \text{Cov} [R_i, R_j] .$$

- ▶ Not clear? Take $x_i = 1/N$ (equally-weighted portfolio)

Diversification and risk: a simple example

Volatility of an equally-weighted portfolio versus the number of stocks



Diversification and risk: summary

Common risk Market-wide:
economic conditions, interest
and exchange rates, inflation, ...
**The asset is affected by the
evolution of the market** (with
its own sensitivity)



Systematic risk =
Undiversifiable risk (market
risk)

Independent risk Specific to the
firm/security i : product market,
quality of the management,
operational risk, ...



Idiosyncratic risk =
Diversifiable risk

Exercise 1: Optimal Portfolio Choice with one risky asset

- ▶ Stock A trades today for a price of EUR 100. Depending on the evolution of the economy, the price of Stock A in one year will be:
 - ▶ 120 with probability 0.5 (expansion)
 - ▶ 100 with probability 0.5 (recession)
- ▶ The risk-free rate is 5%
- ▶ The investment fund AlwaysMore claims to offer an investment product which is superior to Stock A. For an initial investment of 220, it commits itself to pay in one year:
 - ▶ 320 in the expansion
 - ▶ 200 in the recession.
- ▶ Should you accept this offer? If not, what is the highest price you are willing to pay for this investment?

Exercise 2: Optimal Portfolio Choice with two risky assets

- You want to form a portfolio with two stocks A and B. These stocks have expected returns of $E_A = 10\%$ and $E_B = 15\%$ with standard deviations of $\sigma_A = 15\%$ and $\sigma_B = 25\%$. The correlation coefficient between the two stocks is -0.3 .
1. What is the risk and return of a portfolio invested to 40% in A and 60% in B?
 2. If you want to achieve an expected return of 11%, how much risk do you need to take?
 3. Suppose that you can invest in the two stocks and also invest and borrow at the risk-free rate of 5%. Show that you can construct a portfolio with the same return than stock A but with a lower risk.

Exercise (diversification)

- ▶ You can invest in two stocks with the following characteristics

Stock	Expected Return	St.Dev.
A	12%	15%
B	20%	45%

- ▶ What is the risk and the return of a portfolio invested to 50% in A and 50% in B if the correlation coefficient is 0.2?
- ▶ If the stocks are perfectly negatively correlated, what is the portfolio with the lowest risk?

Quiz

- ▶ Define the leverage effect
- ▶ Define the diversification effect
- ▶ Is it possible to augment the return of a portfolio with the risk-free asset?
 - ▶ What happens then to the risk of the portfolio?
- ▶ If the stocks are not perfectly correlated, then the risk of the portfolio is always: a) lower, b) larger c) equal to the weighted average of the individual risks?

Quiz

- ▶ Is it possible to fix the level of risk of a portfolio and how?
- ▶ Is it possible to fix the level of expected return of a portfolio and how?
- ▶ In which case can it be advantageous to buy a stock with higher risk and lower expected return?
- ▶ What are short sales?
- ▶ What are you betting on in the case of a short sale?
- ▶ What does a short position correspond to in terms of portfolio weight?